BundleFuse: Mathematical Definitions of the Raw Scores

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Take a candidate item c and a nonempty observed item set S, with $c \notin S$. Each raw score $r_h(c \mid S)$ supplies one number for that candidate. This document defines these numbers *before* catalogue standardization, validation-selected weighting, and final interpolation. It covers the eight dataset configurations and distinguishes their formulas whenever the same score name has different meanings. All inputs and intermediate quantities used in the equations are defined here.

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1 Shared notation and inputs

1.1 Items, bundles, and incidence

Let $\mathcal{I} = \{1, \dots, n\}$ be the item catalogue and $\mathcal{B} = \{b_1, \dots, b_m\}$ the training bundles. Repeated copies of an item inside one bundle are removed in the graph construction. Define

$$B \in \{0,1\}^{m \times n}, \qquad B_{bi} = \mathbf{1}\{i \in b\}, \qquad x_S \in \{0,1\}^n, \quad (x_S)_i = \mathbf{1}\{i \in S\}.$$

Here b indexes a training bundle, i an arbitrary item, s, t observed items, and c the candidate. Every formula below uses the unit-weight bundles used by the dataset pipelines. A separately supplied bundle-weight option in the graph class is outside this document's scope.

Define the training frequency and training-bundle size by

$$f_i = \sum_b B_{bi}, \qquad d_b = \sum_i B_{bi}.$$

Let α and γ be the frequency and size exponents, respectively, and β the additional-overlap coefficient. Their default values are $\alpha = \gamma = 0.5$ and $\beta = 1.5$. Define

$$a_i = (f_i + 5)^{-\alpha}, \quad v_b = (d_b + 1)^{-\gamma}, \quad D = \operatorname{diag}(a_1, \dots, a_n), \quad V = \operatorname{diag}(v_1, \dots, v_m).$$

Thus D is an $n \times n$ diagonal matrix of item-frequency weights; V is an $m \times m$ diagonal matrix of bundle-size weights. The offsets 5 and 1 are fixed implementation constants, not counts estimated from the query.

For a scalar z, $[z]_+ = \max(z, 0)$. The operation $\mathrm{off}(A)$ replaces the diagonal of a square matrix A by zero. For any matrix F of item row vectors, define row normalization by

$$[\mathcal{N}(F)]_i = \frac{F_i}{\|F_i\|_2 + \epsilon_N}, \qquad \epsilon_N = 10^{-8}.$$

A zero row remains zero. Norm stabilizers make the normalized length slightly less than one. A sum over S and a mean over S are kept distinct throughout.

1.2 Category labels and description arrays

Let $\ell_i \in \{1, \ldots, q\}$ be the primary label of item i. For POG this is the released clothing category; for Spotify it is the artist; for dunnhumby it is the product's sub-commodity label. For electronic, food and clothing it identifies the full parsed category path. Those three datasets also use a second label $\ell_i^{(2)}$ for the first at most two entries of that path. Missing retail/commerce categories receive separate item-specific labels. If explicit primary labels are absent in the general model, they are instead obtained by clustering the available description arrays (or collaborative vectors if no description arrays exist).

Let $X^{(j)} \in \mathbb{R}^{n \times p_j}$ denote the supplied item array for modality j, with p_j coordinates per item. POG provides title, image and description arrays; Spotify provides an audio array. These are loaded from `title_emb.npy`, `content_feature.pt` and `description_feature.pt` as applicable. The scorer consumes these released numbers; it does not infer their originating encoder from the array values. Each modality is row-normalized before the constructions below. Retail and commerce text features are defined explicitly in Section 5.4.

2 Scores from exact training-item membership

These scores use item identities and the bundles in which they occur. First count candidate–seed occurrences; then distinguish normalized direct support, support from multiple seeds in one historical bundle, and propagated support.

2.1 Item co-occurrence: `cooc`

Define

$$C = \mathrm{off}(B^\top B), \qquad C_{ci} = \begin{cases} \sum_b B_{bc} B_{bi}, & c \neq i, \\ 0, & c = i. \end{cases}$$

The raw score is

$$r_{\mathrm{cooc}}(c \mid S) = (C x_S)_c = \sum_{s \in S} C_{cs}. \tag{1}$$

Each historical occurrence of a candidate together with a seed contributes one. A bundle containing several seeds contributes once for each such seed.

2.2 Frequency-weighted direct score: `pair`

For each training bundle define its weighted overlap and overlap count:

$$o_b(S) = \sum_{s \in S} B_{bs} a_s, \qquad k_b(S) = \sum_{s \in S} B_{bs} = |b \cap S|.$$

The direct score is

$$p_c(S) = r_{\text{pair}}(c \mid S) = \sum_b B_{bc} v_b o_b(S) = \left[B^\top V B D x_S\right]_c. \tag{2}$$

Relative to ordinary co-occurrence, rarer seeds have larger a_s, and larger historical bundles have smaller v_b. There is no candidate-frequency factor a_c in this direct score.

The stored zero-diagonal pair matrix is

$$W = \text{off}(B^\top V B D), \qquad W_{cs} = a_s \sum_b B_{bc} v_b B_{bs} \quad (c \neq s).$$

For eligible candidates $c \notin S$, $(W x_S)_c = p_c(S)$. Nevertheless, the implementation computes the full vector $p = B^\top V B D x_S$, *including* its seed coordinates, before propagation. Substituting $W x_S$ for p inside the propagated score would therefore change that score. Generally W is not symmetric because only its seed-side index receives a_s.

2.3 Additional joint-overlap score: `delta_inc`

The extra contribution is

$$\delta_c(S) = \beta \sum_b B_{bc} v_b o_b(S) [k_b(S) - 1]_+. \tag{3}$$

Its matrix form is

$$\delta(S) = \beta B^\top V \left((B D x_S) \odot [B x_S - \mathbf{1}]_+ \right),$$

where $\odot$ denotes coordinatewise multiplication and $\mathbf{1}$ is the m-dimensional all-ones vector. A training bundle with only one seed adds no extra contribution. A bundle with two seeds multiplies its direct contribution by an additional β; a bundle with three adds 2β times that contribution. This is additional credit for *joint membership in the same historical bundle*, not merely for two independent pair counts.

More explicitly, define the candidate–seed–seed count

$$H_{cst} = \sum_b v_b B_{bc} B_{bs} B_{bt}.$$

Then

$$\delta_c(S) = \beta \sum_{\{s,t\} \subseteq S} (a_s + a_t) H_{cst}, \tag{4}$$

where each unordered pair of distinct seeds is included once. This identity also shows that $\delta_c(S) = 0$ for a single-seed query.

2.4 Direct plus additional joint overlap: `simp`

The complete incidence score is

$$g_c(S) = r_{\text{simp}}(c \mid S) = p_c(S) + \delta_c(S). \tag{5}$$

The symbols g_c and δ_c denote different quantities: the former includes direct support; the latter contains only the extra contribution. The commerce profile replaces p by an augmented direct score defined in Section 7.2.

2.5 Propagated scores: `diff2_pair` and `diff2`

Define the $n \times n$ operator

$$L = \operatorname{off}\Big(DB^\top EBD\Big), \qquad E = \operatorname{diag}\big((d_b+1)^{-1}\big)_{b=1}^{m}. \tag{6}$$

In particular,

$$L_{ci} = a_c a_i \sum_b \frac{B_{bc}B_{bi}}{d_b+1} \ (c \neq i), \qquad L_{cc} = 0.$$

Here both item endpoints receive a frequency weight. L is symmetric and nonnegative, but its rows need not sum to one. It is a normalized adjacency operator, not a graph Laplacian.

The raw propagated scores are

$$r_{\text{diff2_pair}}(c \mid S) = (Lp(S))_c, \tag{7}$$

$$r_{\text{diff2}}(c \mid S) = (Lg(S))_c = (Lp(S))_c + (L\delta(S))_c. \tag{8}$$

An intermediate item i passes its score p_i or g_i to candidate c with weight L_{ci}. All items, including seeds, participate as intermediates. Because the normalizations in p and L differ, the first expression is not generally $L^2 x_S$. The commerce profile uses a different propagation matrix.

2.6 Popularity: `pop`

$$r_{\text{pop}}(c \mid S) = \log(1+f_c). \tag{9}$$

It depends on candidate training frequency only and is identical for all queries.

3 Association strengths and their compressed profiles

Co-occurrence counts favor frequent items. The next score compares a pair count with a product of marginal counts. SVD then compresses an item's entire association row, so candidates can be compared through similar patterns of association with other items.

3.1 Positive item association: `ppmi`

Using the zero-diagonal C, define

$$q_i = \sum_j C_{ij}, \qquad Q = \sum_{i,j} C_{ij}.$$

The endpoint count q_i differs from bundle frequency f_i: $q_i = \sum_b B_{bi}(d_b - 1)$. The symmetric matrix M is

$$M_{ij} = \begin{cases} \left[\log\left(\dfrac{C_{ij}Q}{(q_i+10^{-9})(q_j+10^{-9})} + 10^{-12}\right)\right]_+, & C_{ij} > 0, \\ 0, & C_{ij} = 0. \end{cases} \tag{10}$$

The name PPMI denotes positive pointwise mutual information. The raw score is

$$r_{\text{ppmi}}(c \mid S) = (Mx_S)_c = \sum_{s \in S} M_{cs}. \tag{11}$$

Only pair counts larger than their marginal-based reference receive positive credit. The logarithm is natural.

3.2 Compressed association-profile similarity: `csvd`

Let $r = \min(r_0, \max(1, n-2))$, where $r_0 = 128$ by default. A truncated singular value decomposition supplies right singular directions $V_r \in \mathbb{R}^{n\times r}$, left singular directions $U_r \in \mathbb{R}^{n\times r}$ and the diagonal matrix of retained singular values $\Sigma_r \in \mathbb{R}^{r\times r}$. Define

$$Z = MV_r \simeq U_r\Sigma_r, \qquad U = \mathcal{N}(Z). \tag{12}$$

The implementation uses a randomized truncated decomposition with random state zero; MV_r is the operational definition of the returned feature matrix. If no associations are stored, it uses zero features. The raw score is

$$r_{\mathrm{csvd}}(c \mid S) = U_c \left(\frac{1}{|S|} \sum_{s\in S} U_s \right)^{\top} = \frac{1}{|S|} \sum_{s\in S} U_c U_s^{\top}. \tag{13}$$

This averages cosine similarities of compressed association profiles, up to the norm stabilizer. It is not the entry (c, s) of the reconstructed matrix $U_r\Sigma_r V_r^{\top}$. Before row normalization the profile Gram matrix is $U_r\Sigma_r^2 U_r^{\top}$.

Even though M is symmetric, it need not be positive semidefinite. SVD retains directions by singular-value magnitude, not by positive eigenvalues alone. Orthogonality concerns the *columns* of singular-vector matrices; individual item *rows* of $U_r\Sigma_r$ do not have equal norms. This is why row normalization is a separate operation.

4 Scores from primary labels and soft feature groups

These constructions share information between different item identities. The hard-label score uses the supplied category or artist; the soft-label score uses clusters of item features. These are different group definitions.

4.1 Counts of labels inside training bundles

Define the bundle–label count matrix

$$T \in \mathbb{R}^{m\times q}, \qquad T_{ba} = \sum_i B_{bi}\mathbf{1}\{\ell_i = a\}.$$

Unlike B, T may contain values greater than one. Let

$$h_a = \sum_b T_{ba}, \qquad H = \sum_a (h_a + 10^{-9}), \qquad A = T^{\top}T.$$

Here $A \in \mathbb{R}^{q\times q}$ counts label pairs through

$$A_{at} = \sum_{b=1}^{m} T_{ba}T_{bt}.$$

For example, a training bundle containing two items of label a and one item of a different label t contributes $2 \times 1 = 2$ to A_{at}. Thus this is a count of item combinations across labels, rather than just a count of bundles in which both labels are present.

The symbol $J \in \mathbb{R}^{q\times q}_{\geq 0}$ denotes the label-pair count matrix supplied to the association transform below. It is constructed from T, with the diagonal choice determined by the score:

$$J = \begin{cases} \mathrm{off}(T^{\top}T), & \text{different-label association used by } \texttt{cat}, \\ T^{\top}T, & \text{diagonal-retaining commerce } \texttt{homo} \text{ association.} \end{cases}$$

For distinct labels $a \neq t$, both choices have $J_{at} = \sum_b T_{ba}T_{bt}$. For $a = t$, the first choice sets $J_{aa} = 0$; the second retains $J_{aa} = \sum_b T_{ba}^2$. The retained square includes self-item contributions; it is not a count restricted to distinct item pairs. Therefore J is not a new dataset, a learned matrix, or a query-dependent quantity. It is the training label-count product, with or without its diagonal.

For either choice of J, define the transform using the marginals above:

$$[\mathcal{P}_T(J)]_{at} = \begin{cases} \left[\log\left(\dfrac{J_{at}H}{(h_a + 10^{-9})(h_t + 10^{-9})} + 10^{-12}\right)\right]_+, & J_{at} > 0, \\ 0, & J_{at} = 0. \end{cases} \tag{14}$$

The label marginals are occurrence counts in T, not row sums of J. Consequently this transform and the item PPMI construction use different marginals. Their shared logarithmic form does not make their definitions interchangeable.

4.2 Different-label association and slot adjustment: `cat`

Set $P^{\text{cat}} = \mathcal{P}_T(\text{off}(A))$. Its diagonal is zero. Define the unadjusted category score

$$r_{\text{cat}}^{\text{base}}(c \mid S) = \sum_{s \in S} P_{\ell_s, \ell_c}^{\text{cat}}.$$

For the general non-music path, define for each label a

$$n_a^{(1)} = \sum_b \mathbf{1}\{T_{ba} \geq 1\}, \qquad n_a^{(2)} = \sum_b \mathbf{1}\{T_{ba} \geq 2\}, \qquad \eta_a = \frac{n_a^{(2)} + 0.5}{n_a^{(1)} + 1}.$$

The slot adjustment is

$$r_{\text{slot}}(c \mid S) = -\sum_{s \in S} \mathbf{1}\{\ell_c = \ell_s\}\mathbf{1}\{\eta_{\ell_s} < 0.15\}(1 - \eta_{\ell_s}).$$

Then

$$r_{\text{cat}}(c \mid S) = \begin{cases} r_{\text{cat}}^{\text{base}}(c \mid S) + 0.5 r_{\text{slot}}(c \mid S), & \text{general non-music,} \\ r_{\text{cat}}^{\text{base}}(c \mid S), & \text{music or research retail/commerce.} \end{cases} \tag{15}$$

Thus a category seldom repeated within training bundles receives a negative same-category adjustment in the first case. Multiple seeds with that category each contribute the adjustment. Retail/commerce research scores do not apply it.

4.3 Shared-label score: `homo`

Spotify and Spotify sparse. Let $a(i)$ and $b(i)$ be item i's artist and album identifiers, and let N_a and N_b denote the numbers of catalogue items with those identifiers. The raw score is

$$r_{\text{homo}}(c \mid S) = \sum_{s \in S} \left(\frac{\mathbf{1}\{a(c) = a(s)\}}{N_{a(c)} + 5} + 2\frac{\mathbf{1}\{b(c) = b(s)\}}{N_{b(c)} + 2} \right). \tag{16}$$

If album labels are unavailable, omit the album term. These denominators count catalogue items, not training occurrences.

General model with supplied hard group labels. For group type j, let $g_j(i)$ be the supplied label, N_{ja} the number of items in group a, and ω_j its supplied weight. Then

$$r_{\text{homo}}(c \mid S) = \sum_j \omega_j \sum_{s \in S} \frac{\mathbf{1}\{g_j(c) = g_j(s)\}}{N_{j,g_j(c)} + 1}.$$

These groups are supplied labels; they need not be clusters of features.

General non-music model without supplied groups. The fallback definition retains the diagonal of the label count product:

$$r_{\text{homo}}(c \mid S) = \sum_{s \in S} (T^\top T)_{\ell_s, \ell_c}.$$

This is a raw count score, with no logarithmic transform. It is available in the extended profile but outside the POG research selection menu. The commerce research version is defined in Section 7.2.

4.4 Soft feature-group association: `softcat`

Let $F \in \mathbb{R}^{n \times p}$ be the features used for clustering. For POG's research profile, F is the row-normalized concatenation of the normalized title, image and description arrays that are present. Apply MiniBatchKMeans to the item rows, with $k = \max(2, \min(k_0, n-1))$, $k_0 = 400$ by default, random state zero, three initializations and batch size 4096. Let μ_a be cluster a's arithmetic centre and $\widehat{\mu}_a = \mu_a / (\|\mu_a\|_2 + 10^{-8})$. A feature group here is precisely one of these clusters; it is not a training bundle and need not coincide with a clothing category.

With temperature $\tau_s = 0.05$, define soft membership

$$A^{\text{soft}}_{ia} = \frac{\exp(F_i \widehat{\mu}_a^\top / \tau_s - m_i)}{\sum_{t=1}^{k} \exp(F_i \widehat{\mu}_t^\top / \tau_s - m_i) + 10^{-9}}, \qquad m_i = \max_t F_i \widehat{\mu}_t^\top / \tau_s. \tag{17}$$

Each item is assigned a weight for every group rather than only one hard label. Define a separate matrix, $J^{\text{soft}} \in \mathbb{R}^{m \times k}$, by

$$J^{\text{soft}} = B A^{\text{soft}}, \qquad J^{\text{soft}}_{ba} = \sum_{i=1}^{n} B_{bi} A^{\text{soft}}_{ia} = \sum_{i \in b} A^{\text{soft}}_{ia}.$$

Row b describes one training bundle and column a one feature cluster. Its entry is the sum of the membership weights in cluster a of every item in bundle b. For example, membership weights 0.8, 0.3 and 0.1 for three items in that bundle give $J^{\text{soft}}_{ba} = 1.2$. It is a soft group count, not a probability, and can exceed one. This $m \times k$ matrix is distinct from the $q \times q$ hard-label pair matrix J defined in Section 4.1.

The product $((J^{\text{soft}})^\top J^{\text{soft}})_{at} = \sum_b J^{\text{soft}}_{ba} J^{\text{soft}}_{bt}$ measures how much groups a and t are jointly represented across training bundles. Expanding each product also counts a single item's membership in both groups. Remove those same-item contributions, then remove the group diagonal:

$$C^{\text{soft}} = \left[\operatorname{off}\left((J^{\text{soft}})^\top J^{\text{soft}} - \sum_i f_i (A^{\text{soft}}_i)^\top A^{\text{soft}}_i \right) \right]_+. \tag{18}$$

For distinct groups a, t, this equals $\sum_b \sum_{i \neq j} B_{bi} B_{bj} A^{\text{soft}}_{ia} A^{\text{soft}}_{jt}$, apart from numerical clipping. The diagonal is then excluded by construction. Define *catalogue* group masses $u_a = \sum_i A^{\text{soft}}_{ia}$ and $U_0 = \sum_a u_a$, and set

$$P^{\text{soft}}_{at} = \left[\log\left(\frac{C^{\text{soft}}_{at} U_0}{u_a u_t + 10^{-9}} + 10^{-12} \right) \right]_+. \tag{19}$$

The score is

$$r_{\text{softcat}}(c \mid S) = A_c^{\text{soft}} P^{\text{soft}} \left(\sum_{s \in S} A_s^{\text{soft}} \right)^{\top}. \tag{20}$$

The group co-occurrences are training-bundle counts while their denominator uses catalogue group masses. This is the implemented association transform; it should not be silently replaced by a probability model with different marginals. This score is constructed for the non-music general path when description features are available.

5 Description, residual-description, and bridge scores

All dense-feature scores have one scoring rule. If F is the relevant $n \times p$ feature matrix, define

$$r_F(c \mid S) = F_c \left(\frac{1}{|S|} \sum_{s \in S} F_s \right)^{\top}. \tag{21}$$

The observed mean is not normalized again. Different score names specify different constructions of F.

5.1 Direct description similarity: `content`

For the general model,

$$F^{\text{content}} = \mathcal{N}\Big([X^{(1)} \mid \cdots \mid X^{(J)}] \Big), \qquad r_{\text{content}} = r_{F^{\text{content}}}.$$

The vertical separators mean concatenate coordinates, not sum modalities. For research retail/commerce, the content matrix is the normalized 128-dimensional text representation in Section 5.4.

5.2 Within-group residual descriptions

Define $g(i)$ to be the label used for centring and $\mathcal{G}_a = \{i : g(i) = a\}$. For modality j, compute

$$\overline{X}_a^{(j)} = \frac{1}{|\mathcal{G}_a|} \sum_{i \in \mathcal{G}_a} X_i^{(j)}, \qquad R_i^{(j)} = X_i^{(j)} - \overline{X}_{g(i)}^{(j)}, \qquad F^{(j)} = \mathcal{N}(R^{(j)}). \tag{22}$$

The centre is this coordinatewise arithmetic mean over exactly the specified item group, not a mean over all features or over the query. For non-music items, $g(i) = \ell_i$: the supplied clothing category in POG. For music, $g(i)$ comes from a separate MiniBatchKMeans fit on the provided clustering features (concatenated audio features in the released data). Music centring therefore uses feature clusters, not artists.

Raw score key	Modality j	Raw value
`style`	title/general content	$r_{F^{(j)}}(c \mid S)$
`imgstyle`	image	$r_{F^{(j)}}(c \mid S)$
`descstyle`	description	$r_{F^{(j)}}(c \mid S)$
`audiostyle`	audio	$r_{F^{(j)}}(c \mid S)$

Subtracting the group mean removes the shared mean description before comparing the remaining coordinates. It does not prove statistical independence from the group, and row normalization need not preserve a zero group mean. The POG research menu selects `style` and `imgstyle`; description residuals can still enter its bridge even though `descstyle` is not a standalone menu entry.

5.3 History-propagated description similarity: `bridge`

In the general non-music path define

$$F^0 = \mathcal{N}([F^{(1)} \mid \cdots \mid F^{(J)}]).$$

For the Spotify research profile, $F^0 = F^{\text{audio}}$. Using L from Equation (6), construct

$$N = \mathcal{N}(LF^0), \qquad F^{\text{bridge}} = \mathcal{N}(F^0 + \lambda_B N), \qquad r_{\text{bridge}} = r_{F^{\text{bridge}}}, \quad \lambda_B = 0.5. \tag{23}$$

The row $(LF^0)_c = \sum_i L_{ci} F_i^0$ is the sum of descriptions of other items, weighted by their normalized historical co-membership with candidate c. Mixing it with F_c^0 gives the bridge vector. Both the candidate and each seed use these mixed vectors.

For research dunnhumby, start from normalized text content F^{content} instead of category-centred residuals:

$$F^{\text{bridge}} = \mathcal{N}\big(F^{\text{content}} + 0.5\mathcal{N}(LF^{\text{content}})\big).$$

The commerce bridge has a separate sparse-text formula below.

5.4 Text features and the `tfidf` score

For dunnhumby concatenate department, commodity, sub-commodity and brand text. For commerce concatenate item title and its parsed category path. Tokenization uses the vectorizer's default word tokens, lowercasing, English stop-word removal, unigrams and bigrams. Retain at most 8000 terms for dunnhumby and 12000 for the commerce research profile. Vocabulary truncation is based on corpus term frequency after the vectorizer's filtering.

Let t_{iw} be term w's count in item i and $\text{df}_w = \sum_i \mathbf{1}\{t_{iw} > 0\}$. The pre-normalization entry is

$$\widetilde{H}_{iw} = \begin{cases} (1 + \log t_{iw}) \left(1 + \log \frac{1+n}{1+\text{df}_w}\right), & t_{iw} > 0, \\ 0, & t_{iw} = 0. \end{cases}$$

Let H be its row-unit-normalized sparse matrix, with zero rows kept zero. The TF-IDF score is

$$r_{\text{tfidf}}(c \mid S) = (HH^\top x_S)_c = \sum_{s \in S} H_c H_s^\top. \tag{24}$$

This uses a *sum*, not the mean in Equation (21). For the dense content matrix used by the research basket profiles, compute

$$F^{\text{content}} = \mathcal{N}(HV_{r_T}), \qquad r_T = \min(128, \#\text{terms} - 1),$$

where V_{r_T} are the retained right singular directions of H. Vocabulary and document-frequency counts use the catalogue item texts; they do not use hidden query labels. The retail research menu does not select `tfidf`, although its scorer makes this score available as an auxiliary.

6 Candidate support across multiple seeds: `rao`

This score starts from candidate–seed affinities and then multiplies support from distinct seeds. It is different from the historical triple count in δ: it can be nonzero when candidate–seed pairs occurred in different training bundles.

Let F^R be the row-normalized features used in this score. They are title residuals for POG, audio residuals for Spotify, dense bridge features for research dunnhumby, and dense commerce bridge features for research commerce. Let $\zeta = 0.15$, except $\zeta = 0.25$ for research commerce. Define

$$w_c(s) = W_{cs} + \zeta \left[F_c^R (F_s^R)^\top \right]_+, \qquad A_c(S) = \sum_{s \in S} w_c(s). \tag{25}$$

Here W is the zero-diagonal direct-pair matrix already defined. The additional term is positive description similarity; it is not a validation-fitted candidate–seed weight.

When both descriptions and primary labels are available, define

$$G_{st} = 0.5\mathbf{1}\{\ell_s = \ell_t\} + 0.5 \left[F_s^R (F_t^R)^\top \right]_+ \quad (s \neq t), \qquad G_{ss} = 1.$$

With only descriptions, use their positive dot product; with only labels, use the equal-label indicator. With neither, set G to the identity matrix. The seed-pair coefficient is

$$K_{st} = \begin{cases} 0, & s = t, \\ [1 + \gamma_R G_{st}]_+, & s \neq t,\ \gamma_R < 0, \\ 1 + \gamma_R G_{st}, & s \neq t,\ \gamma_R \geq 0. \end{cases} \tag{26}$$

The dataset rule is $\gamma_R = 1$ for Spotify variants and $\gamma_R = -1$ for POG, retail and commerce. The raw score is

$$r_{\mathrm{rao}}(c \mid S) = \frac{\displaystyle\sum_{s \in S} \sum_{t \in S,\ t \neq s} w_c(s) K_{st} w_c(t)}{A_c(S) + 0.05}. \tag{27}$$

The numerator counts ordered pairs, so a symmetric pair contributes twice. For POG, two similar seeds have smaller K_{st} and supply less extra credit; this does not subtract from their ordinary direct scores. For music the sign is reversed: similar seeds have larger K_{st}. Accordingly the music score should not be described as a diversity penalty. A single seed always gives zero.

The graph class also exposes a simpler auxiliary operator using $w_c(s) = W_{cs}$ and $K_{st} = 1$ for $s \neq t$:

$$r_{e_2}(c \mid S) = \frac{\left[\left(\sum_s w_c(s)\right)^2 - \sum_s w_c(s)^2 \right]_+}{\sum_s w_c(s) + 0.05}. \tag{28}$$

It is not a separate default selection-menu score. The legacy combined graph helper can add $0.35 r_{e_2}$ to $p + \delta$; the main BundleFuse scoring path does not make that addition implicitly.

7 Household, user, session, and commerce variants

7.1 Retail household co-occurrence: `hh_cooc`

Let Y_{ui} be the number of training baskets of household u that contain item i. Counts are retained across baskets. Define

$$d_u^Y = \sum_i Y_{ui}, \qquad C^{\mathrm{hh}} = \mathrm{off}\left(Y^\top \mathrm{diag}((d_u^Y + 1)^{-1}) Y \right).$$

The research dunnhumby score is

$$r_{\text{hh_cooc}}(c \mid S) = (C^{\text{hh}} D x_S)_c. \tag{29}$$

The household normalization reduces contributions from households with larger total histories. The seed side receives the same a_s weight as direct bundle support. Same-household items can contribute even when they occurred in different baskets.

7.2 Commerce research profile

This subsection applies to electronic, food and clothing with their default research profile. It modifies several shared scores.

User, session, and pretraining matrices. Let Y^u_{ui} count occurrences of item i across the training bundles assigned to user u, and Y^s_{vi} count occurrences across training bundles assigned to session v. Ignore bundles without a mapped user/session for that corresponding matrix. Define

$$C^u = \text{off}((Y^u)^\top Y^u), \qquad C^s = \text{off}((Y^s)^\top Y^s).$$

Let Y^{pre}_{ui} count rows of the supplied user–item pretraining file for user u, item i. Its third column is ignored. Let $d^{\text{pre}}_u = \sum_i Y^{\text{pre}}_{ui}$. Define

$$C^{\text{pre}} = \text{off}\left((Y^{\text{pre}})^\top \text{diag}\left(\max(d^{\text{pre}}_u, 1)^{-1}\right) Y^{\text{pre}}\right).$$

Then the raw `pretrain` score is

$$r_{\text{pretrain}}(c \mid S) = (C^{\text{pre}} x_S)_c. \tag{30}$$

Augmented incidence and propagation. Using the bundle-only direct score p in Equation (2), set

$$p^{\text{com}} = p + 0.35 C^u x_S + 0.35 C^s x_S, \tag{31}$$
$$g^{\text{com}} = p^{\text{com}} + \delta, \tag{32}$$
$$D_1 = \text{diag}((f_i + 1)^{-1/2}), \tag{33}$$
$$P^{\text{com}} = D_1 (C + 0.5 C^s + 0.5 C^u + 0.2 C^{\text{pre}}) D_1. \tag{34}$$

Thus $r_{\text{pair}} = p^{\text{com}}$, $r_{\text{simp}} = g^{\text{com}}$, $r_{\text{diff2_pair}} = P^{\text{com}} p^{\text{com}}$ and $r_{\text{diff2}} = P^{\text{com}} g^{\text{com}}$, evaluated at candidate coordinate c. The extra δ still comes only from training bundles. P^{com} is not row-normalized. Item `cooc`, `ppmi` and `csvd` continue to use the bundle-only graph, not $C^u + C^s + C^{\text{pre}}$.

Two-level category score. Construct $T^{(1)}$ from full-path labels and $T^{(2)}$ from the first at most two path entries. Apply Equation (14) separately with each matrix's own marginals. Define

$$P^{(j)} = \mathcal{P}_{T^{(j)}}((T^{(j)})^\top T^{(j)}), \quad j \in \{1, 2\}.$$

Here the diagonal is *retained*. The commerce `homo` score is

$$r_{\text{homo}}(c \mid S) = \sum_{s \in S} \left(P^{(1)}_{\ell_s, \ell_c} + 0.5 P^{(2)}_{\ell^{(2)}_s, \ell^{(2)}_c} \right). \tag{35}$$

The separate `cat` score uses the off-diagonal transform of $T^{(1)}$ and no slot adjustment.

Sparse text bridge and dense features for concordance. With sparse TF-IDF H and dense normalized text features F^{content}, define

$$H^{\text{bridge}} = \mathcal{N}_0\big(H + 1.5P^{\text{com}}H + 0.5(P^{\text{com}})^2H\big)\,, \tag{36}$$

$$F^R = \mathcal{N}\big(F^{\text{content}} + 1.5P^{\text{com}}F^{\text{content}} + 0.5(P^{\text{com}})^2F^{\text{content}}\big)\,. \tag{37}$$

Here $\mathcal{N}_0$ is exact nonzero-row unit normalization with zero rows preserved, as used by the sparse normalization routine. The commerce raw bridge score is

$$r_{\text{bridge}}(c \mid S) = \big[H^{\text{bridge}}(H^{\text{bridge}})^\top x_S\big]_c. \tag{38}$$

This is a sum of sparse-text similarities, not a mean of dense-vector similarities. F^R is used for `rao`, not for this bridge score.

7.3 General auxiliary incidence scores

For an explicitly supplied context-by-item incidence matrix $I \in \mathbb{R}^{t\times n}$, the general model supports a named score

$$r_{\text{aux}}(c \mid S) = \begin{cases} [I^\top IDx_S]_c, & \text{seed-frequency weighting enabled,} \\ [I^\top Ix_S]_c, & \text{otherwise.} \end{cases} \tag{39}$$

The matrix may already contain context-degree normalization. The general path does not remove the diagonal of $I^\top I$; eligible candidates exclude the seeds. The loader's extended retail `hh_cooc` and commerce `pretrain` incidences use $I_{ui} = Y_{ui}/\sqrt{\sum_j Y_{uj} + 1}$. They are not identical constructions to the research commerce pretraining normalization above.

7.4 Personalized auxiliaries: `user_freq`, `user_cooc`

These are available in the general extended path when a user history is supplied. For query user u, let n_{ui} be the number of occurrences of item i in that supplied history, and $h_{ui} = \mathbf{1}\{n_{ui} > 0\}$. Then

$$r_{\text{user_freq}}(c \mid S, u) = n_{uc}, \tag{40}$$

$$r_{\text{user_cooc}}(c \mid S, u) = (Ch_u)_c. \tag{41}$$

The first retains repetitions; the second uses the distinct-history indicator. Neither uses the current query seeds directly. Unknown or absent users receive zero scores. These are not selected by the default research basket menus.

8 Inventory and the boundary of this document

8.1 Default research selection menus

The following lists identify which raw sources are offered for ordinary validation weight selection. Offered does not mean selected with positive weight in a fitted model.

Dataset	Raw source menu
POG, POG dense	`cat`, `softcat`, `style`, `imgstyle`, `bridge`, `simp`, `diff2`, `pop`, `cooc`, `ppmi`, `csvd`.
Spotify, Spotify sparse	`simp`, `diff2`, `pop`, `cooc`, `ppmi`, `csvd`, `cat`, `homo`, `audiostyle`, `bridge`.
Dunnhumby	`simp`, `diff2`, `pop`, `cooc`, `hh_cooc`, `ppmi`, `csvd`, `cat`, `content`, `bridge`.
Electronic, food, clothing	`simp`, `diff2`, `cooc`, `ppmi`, `csvd`, `cat`, `homo`, `tfidf`, `bridge`, `pretrain`.

The fitting/scoring logic additionally uses `pair`, `diff2_pair` and `rao` to form controlled combinations with these sources. `delta_inc` is directly addressable in the selected general scorer; the ordinary full scorer uses it internally through `simp` and `diff2`. Placeholder entries `contrast` and `img` in the basket reference scorer contain no numerical score and are not menu entries.

8.2 Optional content compression

The default configurations set `content_rank` to null. If optional content compression is enabled with rank r_C, the stored content, style and bridge matrices with more than r_C columns are replaced by FV_{r_C} from a truncated SVD. They are not row-normalized again. Equation (21) then uses these projected matrices; calling the result an exact cosine similarity would be incorrect. The research basket profiles reject this override to preserve their sparse/dense constructions.

8.3 Experimental comparison operator, outside BundleFuse

The novelty experiments contain an amplified-neighbour comparator. It is not an additional BundleFuse raw source. To distinguish it precisely, define

$$t_b(S) = \frac{\sum_{s \in S} B_{bs} a_s}{\sqrt{d_b}\sqrt{|S|}}.$$

Keep all bundles or only the k_N largest t_b, using bundle index to break ties; let $\mathcal{K}(S)$ denote those retained indices. For chosen power p_N, the comparator is

$$r_{\mathrm{amp}}(c \mid S) = \sum_{b \in \mathcal{K}(S)} B_{bc}\, t_b(S)^{p_N}.$$

This is a predecessor-inspired control; its implementation does not claim an exact reproduction of another published method.

8.4 What happens after a raw score

The scalar functions above are the raw inputs to subsequent standardization and fusion. Their fixed construction constants (such as 5, 0.5 and 0.05) are distinct from validation-selected fusion weights. Selecting a zero fusion weight does not change a raw source's definition, but serving can omit its computation when no remaining selected score depends on it.

Implementation trace for verification

This document was derived from executable expressions in the following files, rather than earlier mathematical documents. The mapping is provided only for checking the equations; no external document is required to define a symbol.

- `bundlefuse/operators.py`: graph, incidence surplus, propagation, item PPMI/SVD, category transform, soft groups, residuals and concordance.
- `bundlefuse/model.py` and `bundlefuse/selected_signals.py`: construction of description matrices and assignment of raw score names.
- `bundlefuse/basket_signals.py`: research retail and commerce variants.
- `bundlefuse/datasets/base.py` and `bundlefuse/datasets/baskets.py`: label sources, text arrays and auxiliary incidence construction.
- `experiments/novelty/raw.py`: amplified-neighbour comparison only.

The experimental source files and running experiment configuration were not modified to prepare this document.

For numerical verification, 87 comparisons on a small independent synthetic bundle collection checked the displayed core equations and the full versus selected scoring paths for fashion, music, retail and commerce, including single-seed and multiple-seed queries. The comparisons passed with relative tolerance 3×10^{-5} and absolute tolerance 3×10^{-6}.